Step 5000 Evaluation and Network-Only Jet3 Ablation

2D airfoil antisymmetric NCL, continuous analytic Jet3, K=1024. The fixed checkpoint is optimizer step 5000; no later weights are used.

1. Fixed three-frame reconstruction

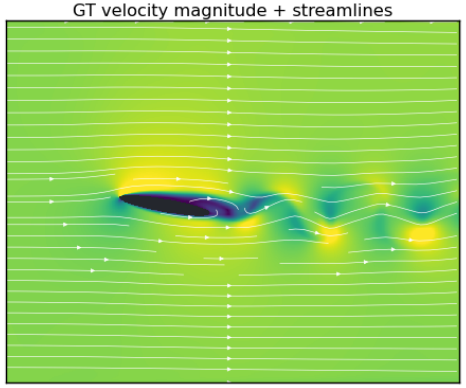
Velocity-L2 coefficient fit; phase 0.5; Ritz quadrature 4096; field batch 32768.

case	alpha	beta	Ritz rank	velocity rel. L2	vorticity rel. L2	wake velocity rel. L2
case 1	+10.0 deg	0.50	1024	2.21%	26.65%	4.74%
case 2	+0.0 deg	0.85	1024	1.40%	16.64%	2.59%
case 3	-10.0 deg	0.50	1024	2.09%	26.45%	4.38%

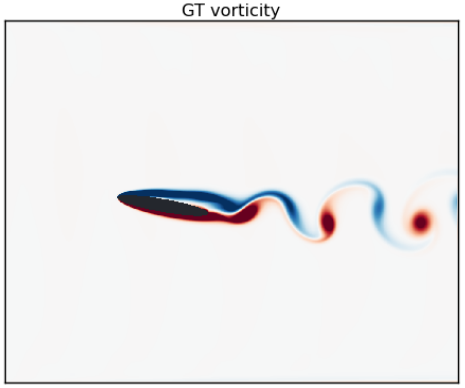
PASS: all full-domain velocity relative L2 errors are below the 10% review threshold. Training remains paused.

Airfoil case: $\beta = 0.500$, $\alpha = +10.00^\circ$

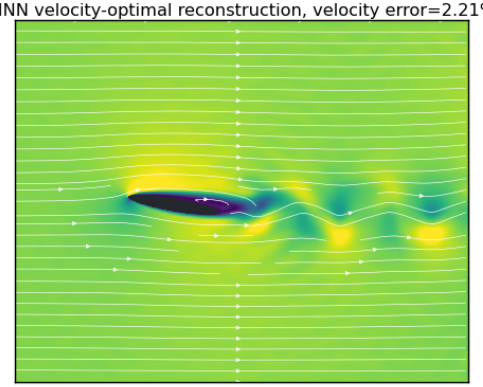
GT velocity magnitude + streamlines



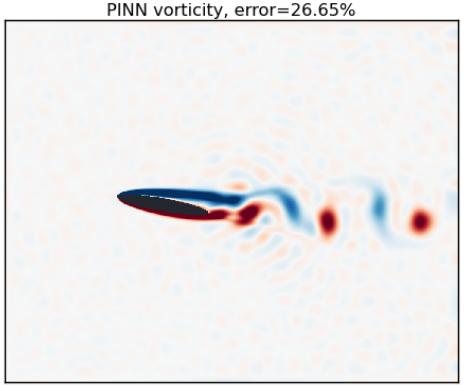
GT vorticity



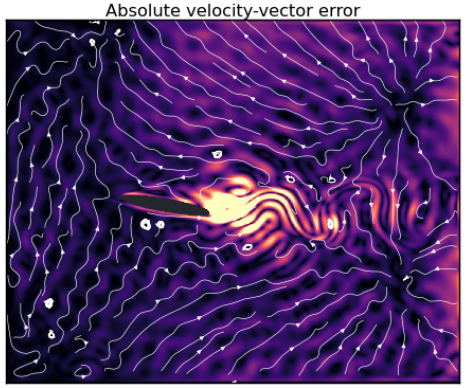
PINN velocity-optimal reconstruction, velocity error=2.21%



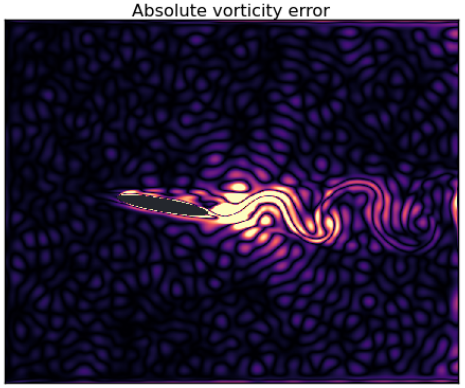
PINN vorticity, error=26.65%



Absolute velocity-vector error

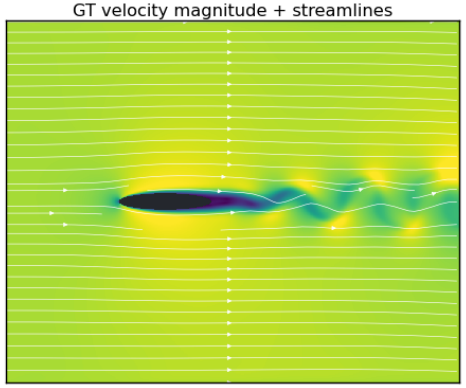


Absolute vorticity error

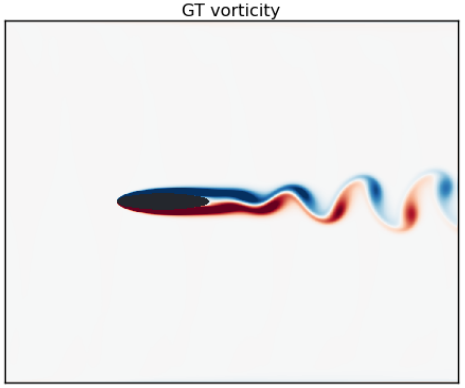


Airfoil case: $\beta = 0.850$, $\alpha = +0.00^\circ$

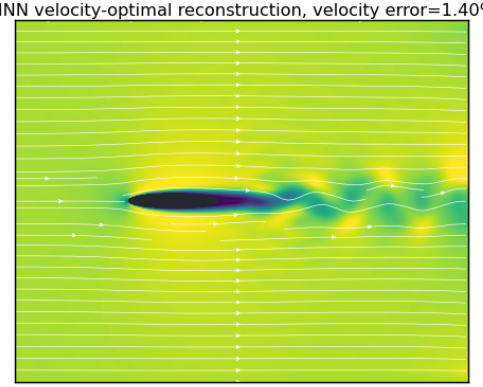
GT velocity magnitude + streamlines



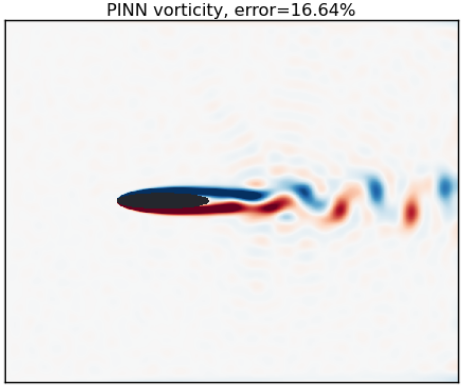
GT vorticity



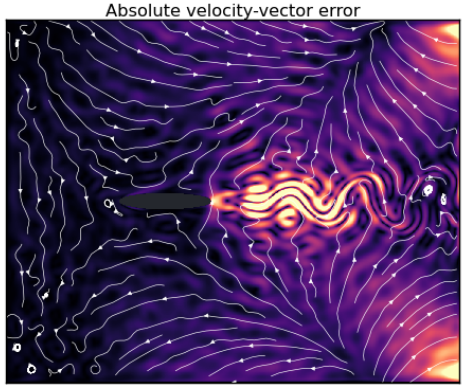
PINN velocity-optimal reconstruction, velocity error=1.40%



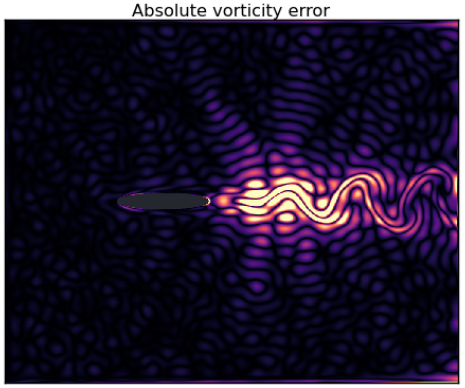
PINN vorticity, error=16.64%



Absolute velocity-vector error

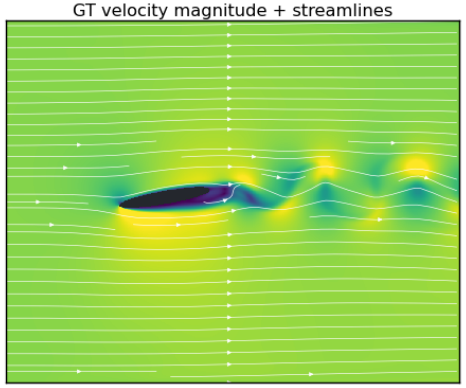


Absolute vorticity error

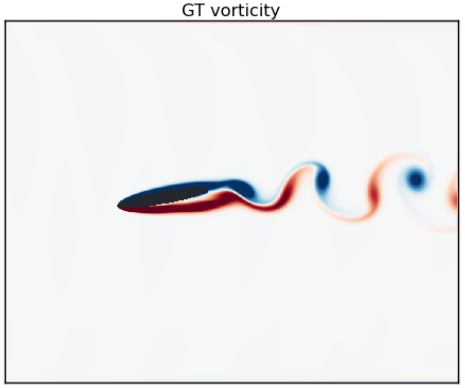


Airfoil case: $\beta = 0.500$, $\alpha = -10.00^\circ$

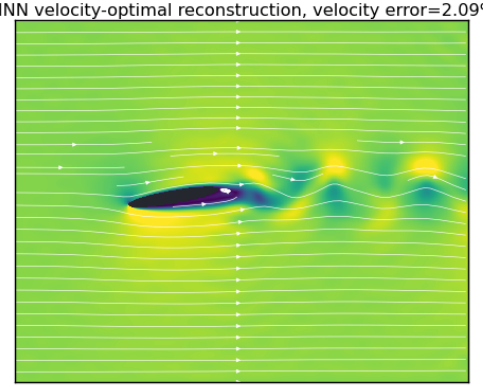
GT velocity magnitude + streamlines



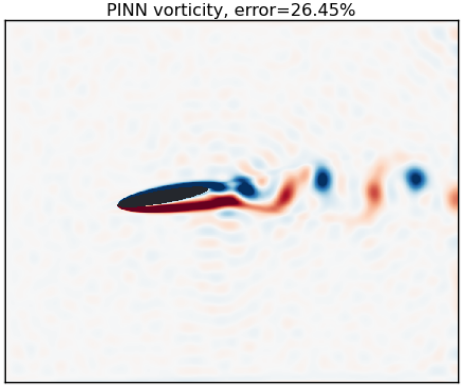
GT vorticity



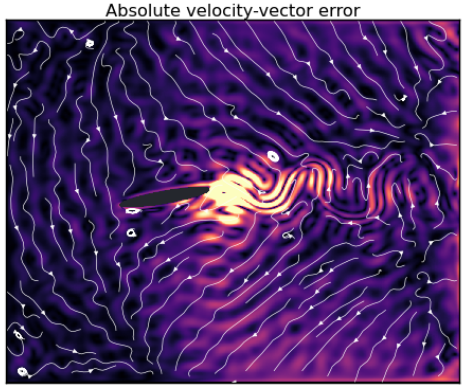
PINN velocity-optimal reconstruction, velocity error=2.09%



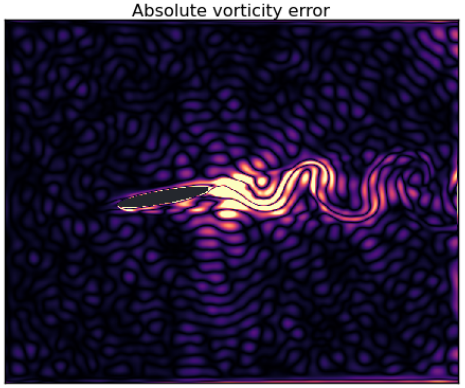
PINN vorticity, error=26.45%



Absolute velocity-vector error

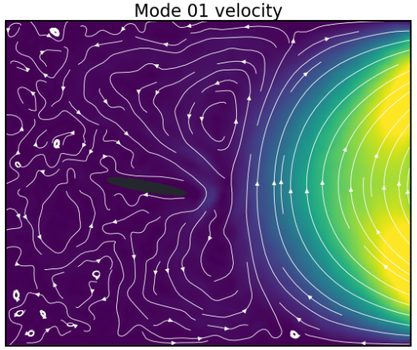


Absolute vorticity error

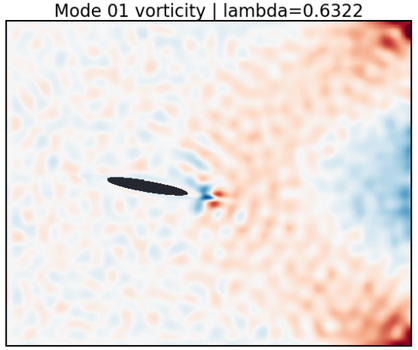


First nine mass-orthogonal Ritz modes | beta=0.50, alpha=+10.0 deg

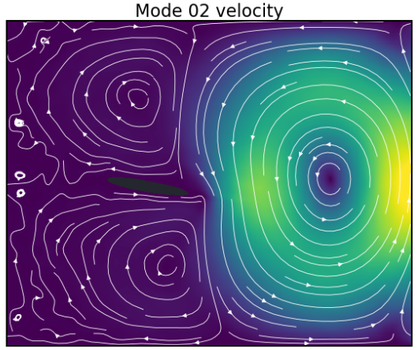
Mode 01 velocity



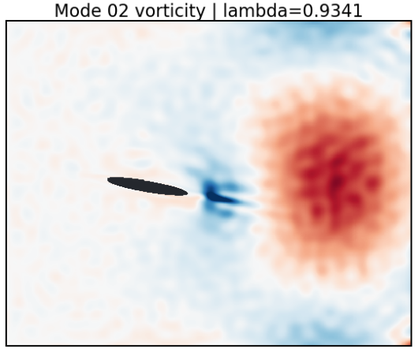
Mode 01 vorticity | lambda=0.6322



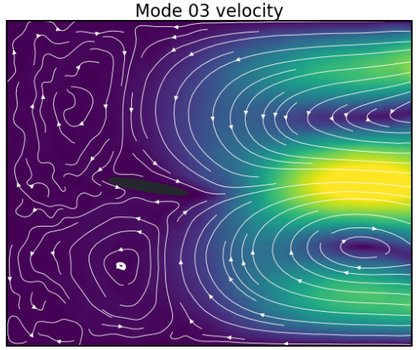
Mode 02 velocity



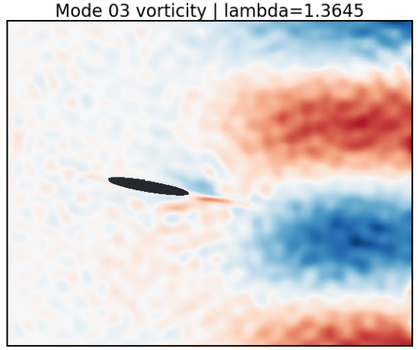
Mode 02 vorticity | lambda=0.9341



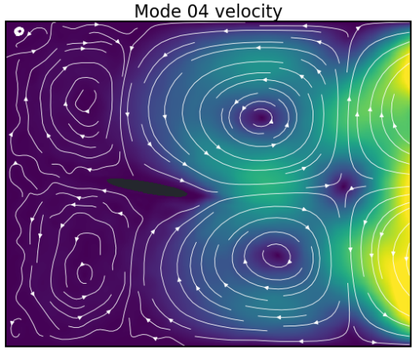
Mode 03 velocity



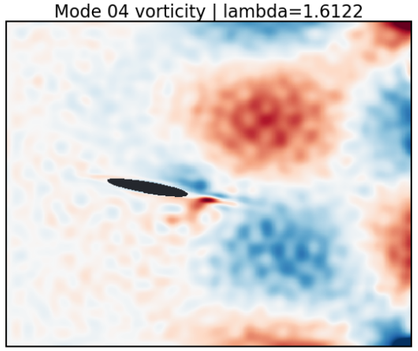
Mode 03 vorticity | lambda=1.3645



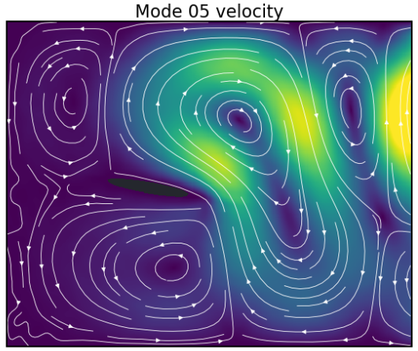
Mode 04 velocity



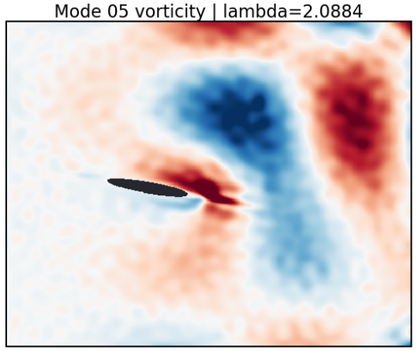
Mode 04 vorticity | lambda=1.6122



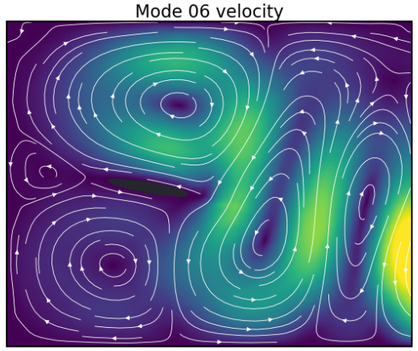
Mode 05 velocity



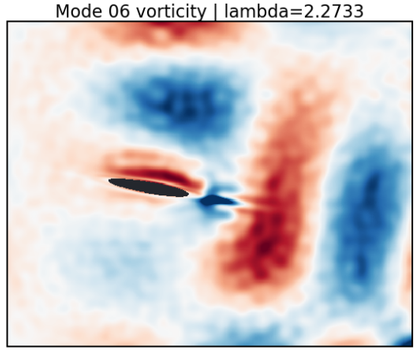
Mode 05 vorticity | lambda=2.0884



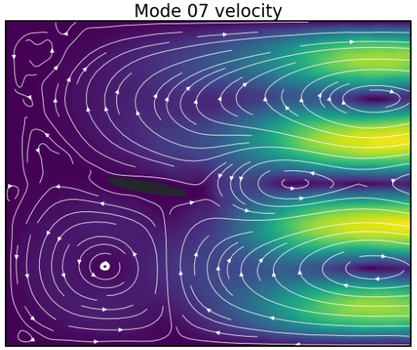
Mode 06 velocity



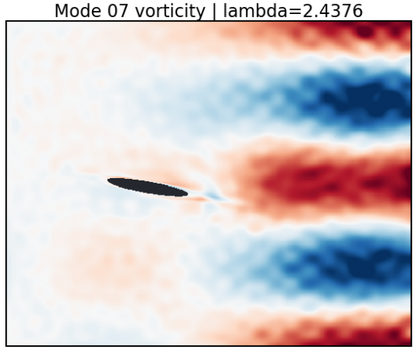
Mode 06 vorticity | lambda=2.2733



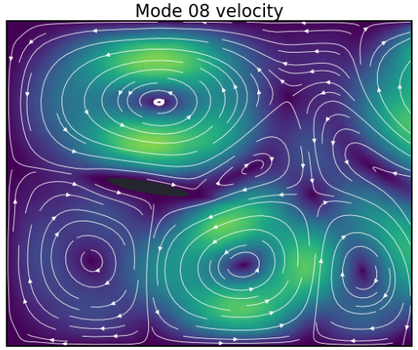
Mode 07 velocity



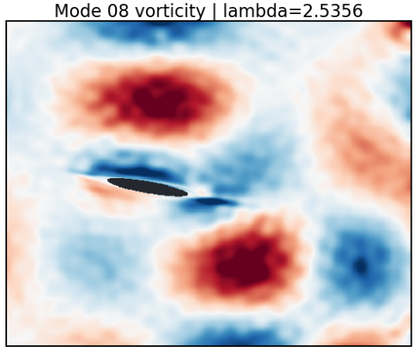
Mode 07 vorticity | lambda=2.4376



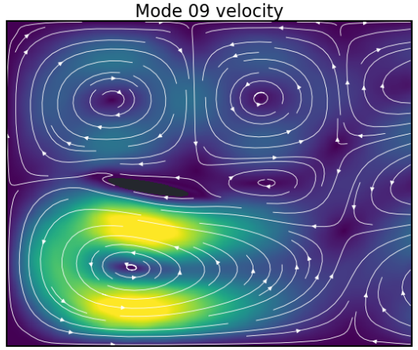
Mode 08 velocity



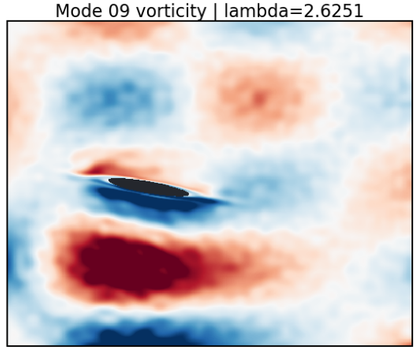
Mode 08 vorticity | lambda=2.5356



Mode 09 velocity

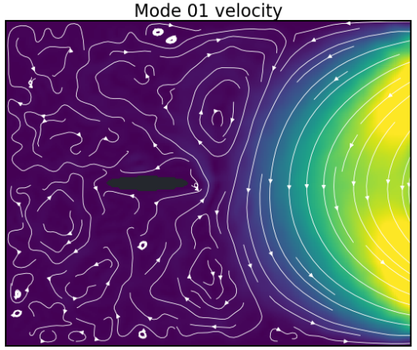


Mode 09 vorticity | lambda=2.6251

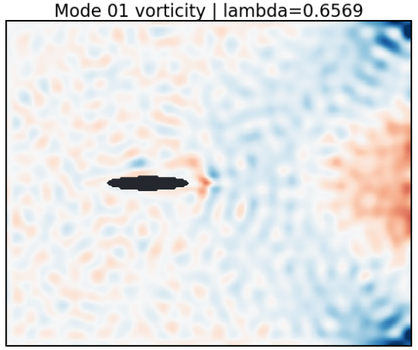


First nine mass-orthogonal Ritz modes | beta=0.85, alpha=+0.0 deg

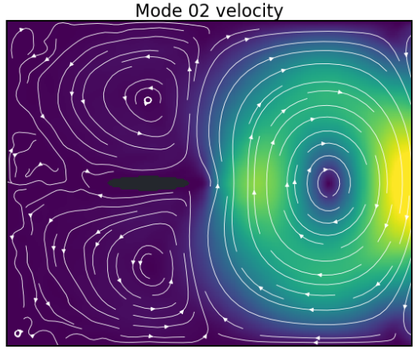
Mode 01 velocity



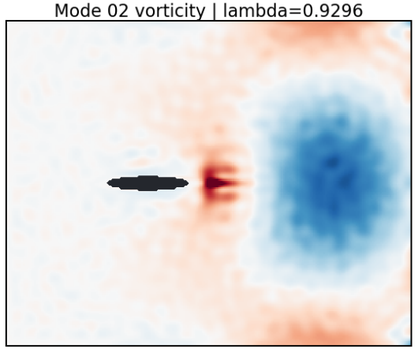
Mode 01 vorticity | lambda=0.6569



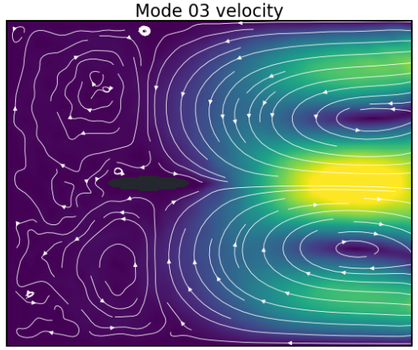
Mode 02 velocity



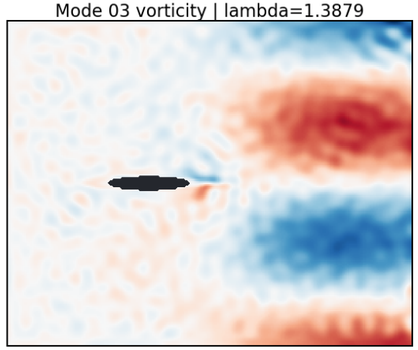
Mode 02 vorticity | lambda=0.9296



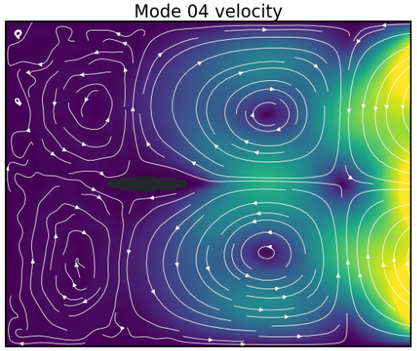
Mode 03 velocity



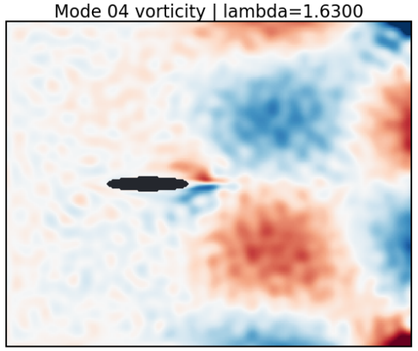
Mode 03 vorticity | lambda=1.3879



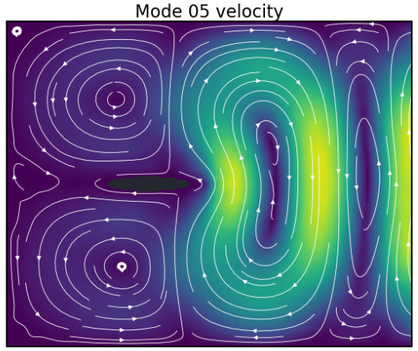
Mode 04 velocity



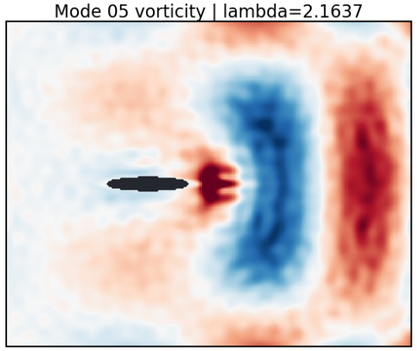
Mode 04 vorticity | lambda=1.6300



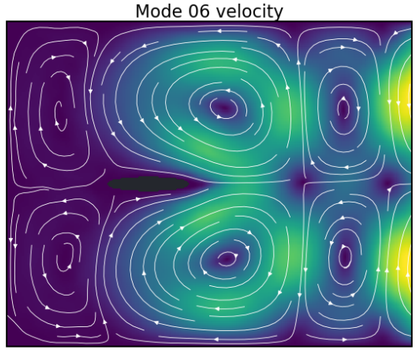
Mode 05 velocity



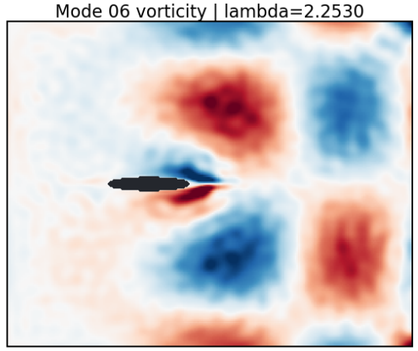
Mode 05 vorticity | lambda=2.1637



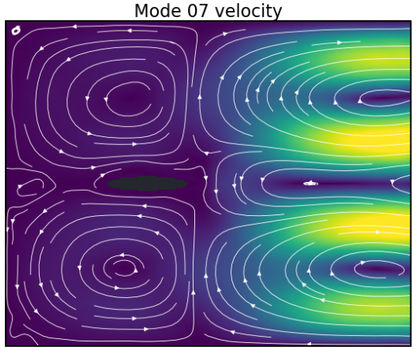
Mode 06 velocity



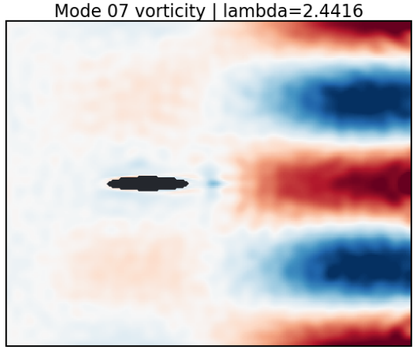
Mode 06 vorticity | lambda=2.2530



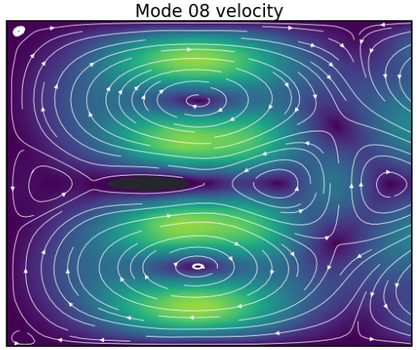
Mode 07 velocity



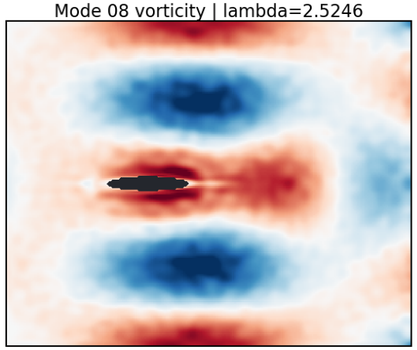
Mode 07 vorticity | lambda=2.4416



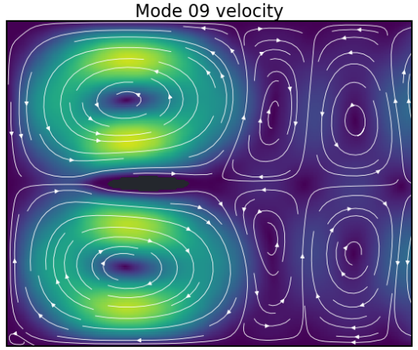
Mode 08 velocity



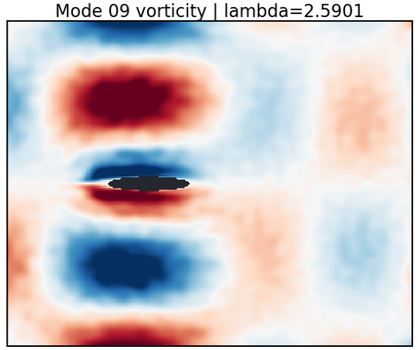
Mode 08 vorticity | lambda=2.5246



Mode 09 velocity

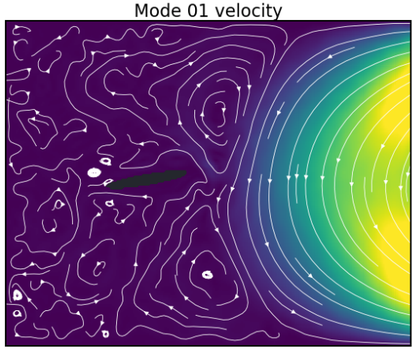


Mode 09 vorticity | lambda=2.5901

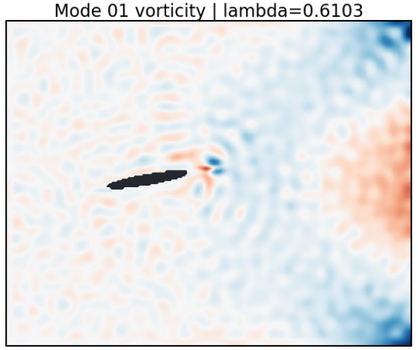


First nine mass-orthogonal Ritz modes | beta=0.50, alpha=-10.0 deg

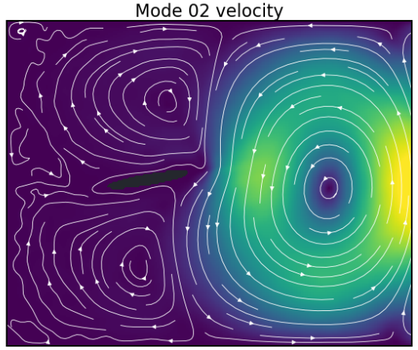
Mode 01 velocity



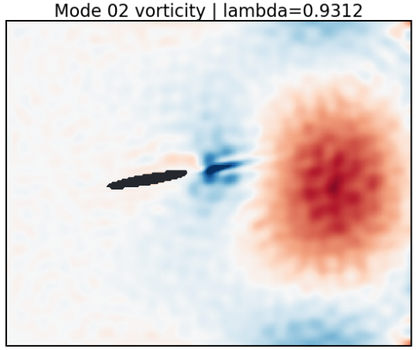
Mode 01 vorticity | lambda=0.6103



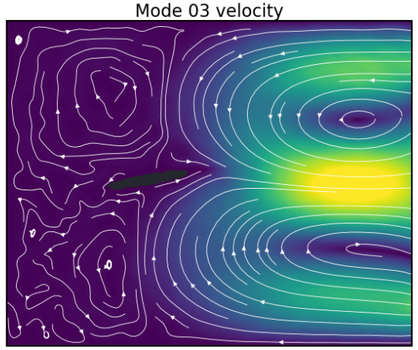
Mode 02 velocity



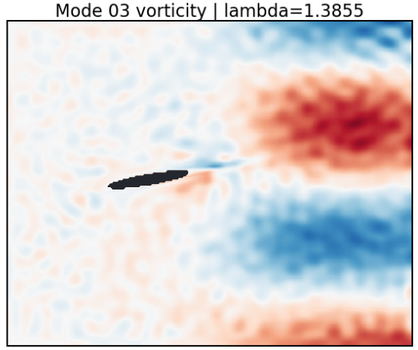
Mode 02 vorticity | lambda=0.9312



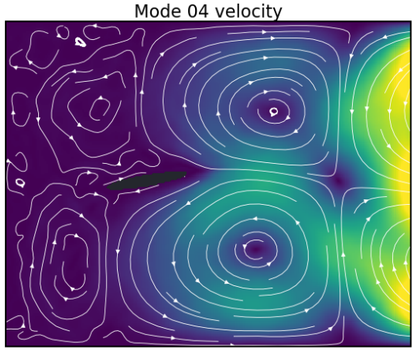
Mode 03 velocity



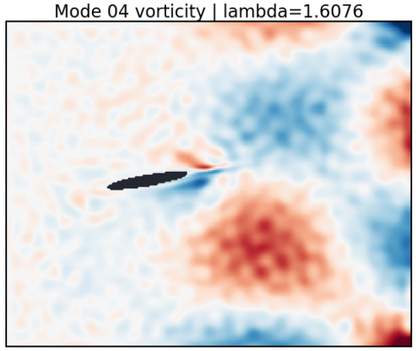
Mode 03 vorticity | lambda=1.3855



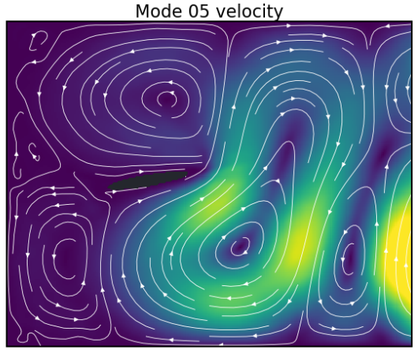
Mode 04 velocity



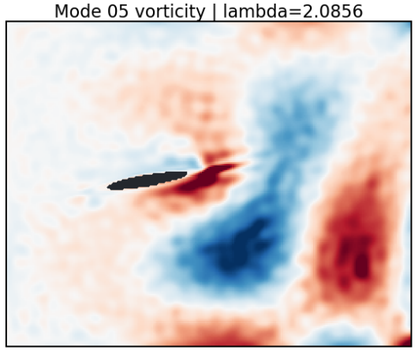
Mode 04 vorticity | lambda=1.6076



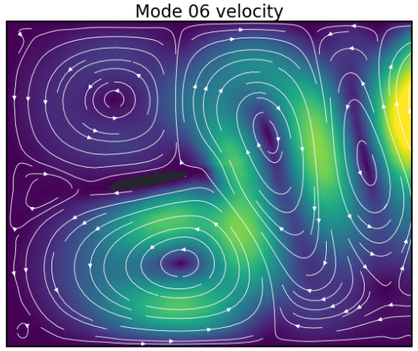
Mode 05 velocity



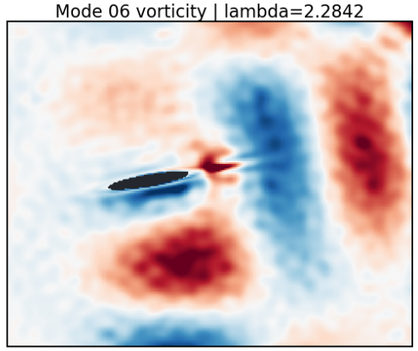
Mode 05 vorticity | lambda=2.0856



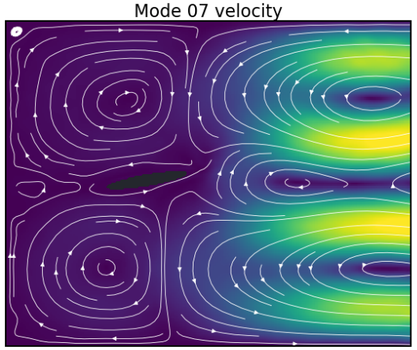
Mode 06 velocity



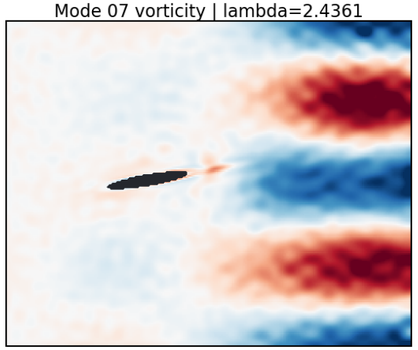
Mode 06 vorticity | lambda=2.2842



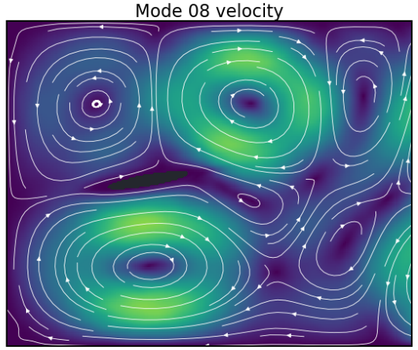
Mode 07 velocity



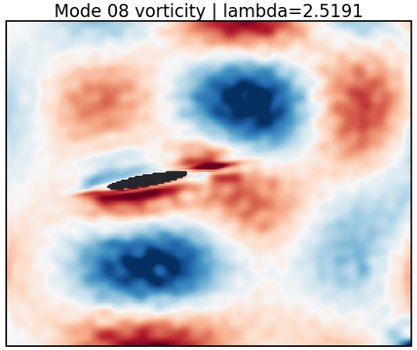
Mode 07 vorticity | lambda=2.4361



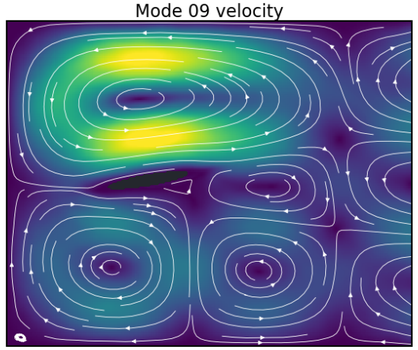
Mode 08 velocity



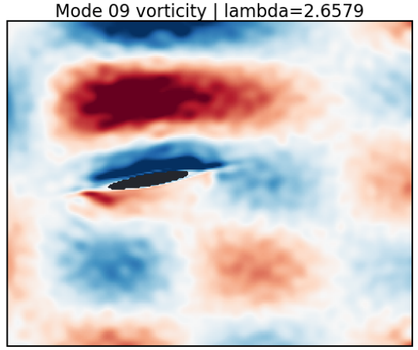
Mode 08 vorticity | lambda=2.5191



Mode 09 velocity



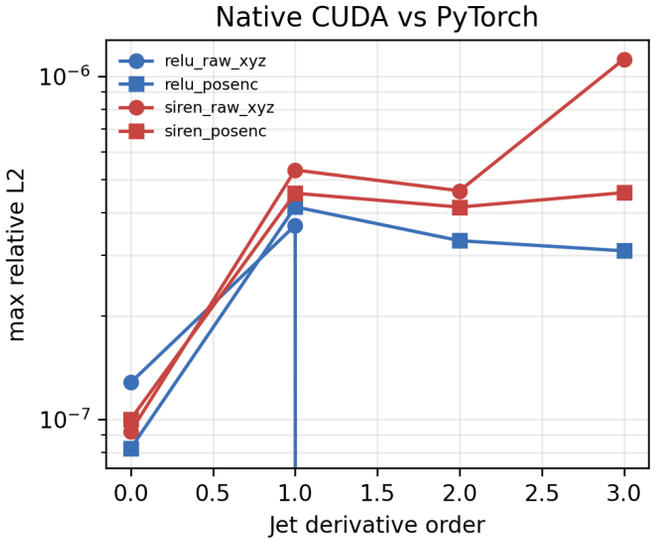
Mode 09 vorticity | lambda=2.6579

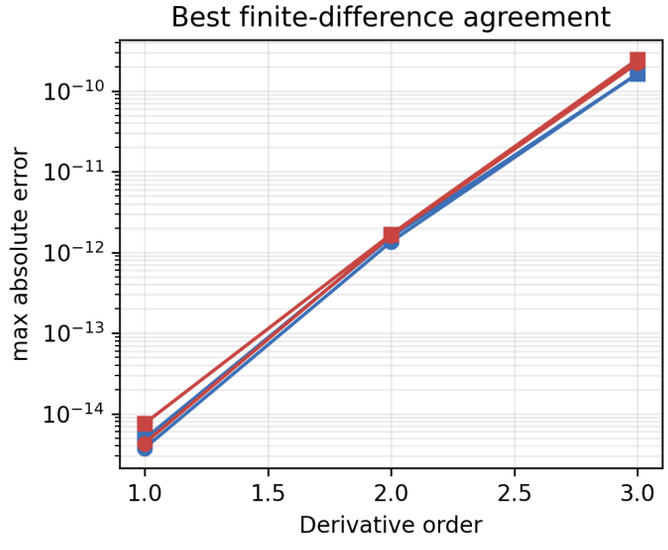


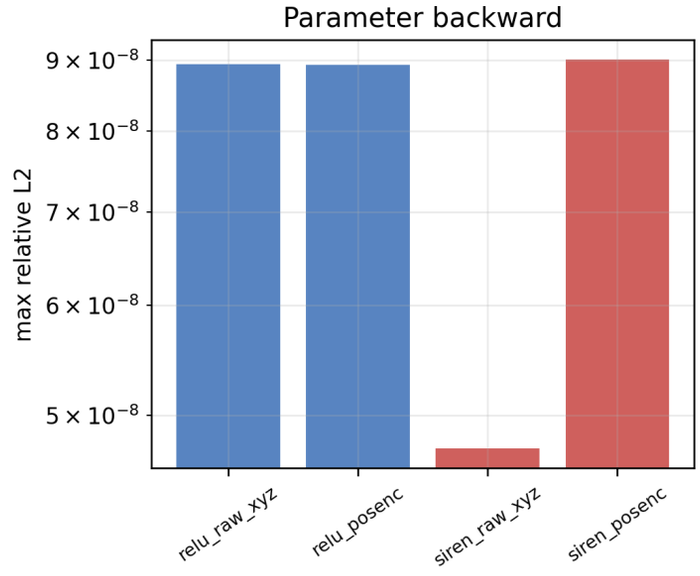
3. Network-only controlled validation

Shared 12-32-32-32-2 topology, FP32 master weights, seed, and coordinates. Native CUDA dense symmetric Jet3 is compared with PyTorch autograd over all 20 slots; parameter backward is checked layer by layer; finite differences use a step-size sweep.

variant	CUDA/AD o0	o1	o2	o3	backward	FD abs o1	FD abs o2	FD abs o3	status
relu_raw_xyz	1.28e-07	3.67e-07	0.00e+00	0.00e+00	8.94e-08	3.72e-15	1.36e-12	1.65e-10	PASS
relu_posenc	8.22e-08	4.16e-07	3.31e-07	3.09e-07	8.92e-08	4.86e-15	1.56e-12	1.63e-10	PASS
siren_raw_xyz	9.20e-08	5.33e-07	4.63e-07	1.12e-06	4.73e-08	4.28e-15	1.62e-12	2.23e-10	PASS
siren_posenc	9.99e-08	4.56e-07	4.15e-07	4.58e-07	9.01e-08	7.54e-15	1.67e-12	2.49e-10	PASS







4. Controlled network effects

These are function changes caused by activation or encoding choices, not implementation errors.

controlled change	normalized output change	max absolute change
activation_at_raw_xyz	0.45828	6.30677e-02
activation_at_posenc	0.45682	6.34498e-02
encoding_at_relu	0.00341	7.67425e-04
encoding_at_siren	0.00595	1.42539e-03

Decision state: the 10% velocity threshold is not exceeded. Training remains paused until an explicit user decision. The checkpoint is not embedded in this report.